



**SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES**



SLOT A

DSA 0404 - Fundamentals of Data Science

K. Lahari

192325118

AI & ML

SIMATS ENGINEERING

1.Find mean, median, mode and range for: 12, 15, 18, 15, 20, 22, 15, 25.

Given data:

12,15,18,15,20,22,15,25

Mean

Formula:

$$\bar{x} = \sum x/n$$

$$\sum x = 12 + 15 + 18 + 15 + 20 + 22 + 15 + 25 = 142$$

$$\bar{x} = \frac{142}{8} = 17.75$$

Median

Arrange the data:

12,15,15,15,18,20,22,25

Since n=8:

$$\begin{aligned}\text{Median} &= 4^{th} + 5^{th} / 2 \\ &= 15 + 18 / 2 = 16.5\end{aligned}$$

Mode

15 occurs most frequently.

$$\text{Mode} = 15$$

Range

Formula:

$$\begin{aligned}\text{Range} &= \text{Maximum} - \text{Minimum} \\ &= 25 - 12 = 13\end{aligned}$$

Answer

- Mean = 17.75
- Median = 16.5
- Mode = 15
- Range = 13

2. Construct a frequency distribution for: 2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Given data:

2,3,4,2,5,3,4,2,6,5,3,4

Value	Frequency
2	3
3	3
4	3
5	2
6	1
Total	12

Therefore, the frequency distribution is:

2:3, 3:3, 4:3, 5:2, 6:1

3. Find Q1, Q2, Q3 and IQR for: 5, 8, 10, 12, 15, 18, 20, 22, 25, 30.

Given data:

5,8,10,12,15,18,20,22,25,30

The data is already arranged in ascending order.

Q2 (Median)

There are 10 values:

$$Q2 = 15 + 18/2 = 16.5$$

Q1

Lower half:

5,8,10,12,15

Middle value:

$$Q1 = 10$$

Q3

Upper half:

18,20,22,25,30

Middle value:

$$Q3=22$$

IQR

Formula:

$$IQR=Q3-Q1$$

$$IQR=22-10=12$$

Final Answer:

- $Q1 = 10$
- $Q2 = 16.5$
- $Q3 = 22$
- $IQR = 12$

4. Detect outliers using the IQR method for: 10, 12, 13, 15, 16, 17, 18, 20, 45.

Given data:

10,12,13,15,16,17,18,20,45

Q1 and Q3

Lower half:

10,12,13,15

$$Q1=12+13/2=12.5$$

Upper half:

17,18,20,45

$$Q3=18+20/2=19$$

IQR

$$IQR=Q3-Q1$$

$$=19-12.5=6.5$$

Outlier Formula

$$\text{Lower Fence} = Q1 - 1.5(IQR)$$

$$= 12.5 - 1.5(6.5)$$

$$= 12.5 - 9.75 = 2.75$$

$$\text{Upper Fence} = Q3 + 1.5(IQR)$$

$$= 19 + 9.75 = 28.75$$

Any value below 2.75 or above 28.75 is an outlier.

Since:

$$45 > 28.75$$

45 is an outlier

Q5. Determine the skewness (Asymmetry) of: 5, 6, 7, 7, 8, 8, 9, 20.

Dataset: 5, 6, 7, 7, 8, 8, 9, 20 (n = 8)

Mean

$$\text{Mean} = \Sigma x / n$$

$$\text{Sum} = 5 + 6 + 7 + 7 + 8 + 8 + 9 + 20 = 70$$

$$\text{Mean} = 70 / 8 = 8.75$$

Median

Sorted data:

5, 6, 7, 7, 8, 8, 9, 20

n=8 (even)

Median = average of 4th & 5th values

$$= (7 + 8) / 2 = 7.5$$

Standard Deviation (population)

$$\sigma = \sqrt{(\Sigma(x - \bar{x})^2 / n)}$$

Squared deviations from mean 8.75:

14.06, 7.56, 3.06, 3.06, 0.56, 0.56, 0.06, 126.56

Sum of squared deviations

$$\sigma = \sqrt{(155.5 / 8)} = \sqrt{19.44} = 4.41$$

Pearson's Coefficient of Skewness

$$\text{Skewness} = 3 \times (\text{Mean} - \text{Median}) / \sigma$$

Skewness:

$$\begin{aligned} & 3 \times (8.75 - 7.5) / 4.41 \\ &= 3 \times 1.25 / 4.41 \\ &= 3.75 / 4.41 \approx 0.85 \end{aligned}$$

Interpretation

- Mean (8.75) > Median (7.5) → distribution is right-skewed (positively skewed)
- Skewness coefficient $\approx +0.85$ confirms moderate positive skew
- The value 20 is a high extreme value pulling the mean to the right

Q6. A uniformly distributed variable ranges from 10 to 50. Find $P(20 < X < 35)$.

X is uniformly distributed on the interval [10, 50], i.e. $X \sim U(10, 50)$

Formula

For $X \sim U(a, b)$: $f(x) = 1 / (b - a)$, $P(c < X < d) = (d - c) / (b - a)$

Here $a = 10$, $b = 50$

So, total width = $b - a = 50 - 10 = 40$

Probability density $f(x) = 1/40$ for $10 \leq x \leq 50$

Required interval width = $35 - 20 = 15$

$$P(20 < X < 35) = 15 / 40 = 0.375$$

Answer: $P(20 < X < 35) = 0.375$ (37.5%)

Q7. A normal distribution has mean 100 and standard deviation 15. Find the Z-score for X=130.

Given: Mean (μ) = 100, Standard Deviation (σ) = 15

Formula

$$\begin{aligned} Z &= (X - \mu) / \sigma \\ Z &= (130 - 100) / 15 \\ &= 30 / 15 \\ &= 2 \end{aligned}$$

Q8. Find P(X < 130) for the Same Normal Distribution

Using $\mu = 100$, $\sigma = 15$, and the Z-score computed in Q7 ($Z = 2$)

Formula

$$\begin{aligned} P(X < x) &= P(Z < (x - \mu)/\sigma) = \Phi(Z) \\ Z &= (130 - 100)/15 = 2 \text{ (from Q7)} \end{aligned}$$

Look up $\Phi(2)$ in the standard normal (Z) table

$$\Phi(2) = 0.9772$$

Answer: $P(X < 130) = 0.9772$ ($\approx 97.72\%$)

Q9. Given a dataset with one extreme value, compare the mean and median before and after outlier treatment.

Sample dataset with an extreme value: 10, 12, 13, 14, 15, 100

Before Outlier Treatment

Mean:

$$\text{Mean} = \Sigma x / n$$

Sum = 10+12+13+14+15+100 = 164, n = 6

$$\text{Mean} = 164/6 = 27.33$$

Median:

Median = middle value(s) of sorted data

Sorted:

10,12,13,14,15,100

$$\text{Median} = \text{avg}(13,14) = 13.5$$

After Removing the Outlier (100)

Treated dataset: 10, 12, 13, 14, 15 (n = 5)

Mean:

$$\text{Sum} = 10+12+13+14+15 = 64$$

$$\text{Mean} = 64/5 = 12.8$$

Median:

$$\text{Median (middle of 5 values)} = 13$$

Comparison Table

Measure	Before Treatment	After Treatment	Change
Mean	27.33	12.80	Drops sharply by 14.53
Median	13.5	13.0	Changes only slightly by 0.5

Conclusion

- The mean is highly sensitive to extreme values (pulled strongly toward the outlier)
- The median is robust/resistant to outliers and barely changes
- For skewed data or data with outliers, the median is a more reliable measure of central tendency

Q10. Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers and skewness.

Dataset (marks out of 100, sorted): 45, 50, 55, 60, 62, 65, 70, 75, 80, 95 (n = 10)

1. Central Tendency

Mean:

$$\text{Mean} = \Sigma x/n$$

$$\text{Sum} = 45+50+55+60+62+65+70+75+80+95 = 657$$

$$\text{Mean} = 657/10 = 65.7$$

Median:

$$\begin{aligned}\text{Median} &= \text{average of 5th \& 6th values} \\ &= (62+65)/2 = 63.5\end{aligned}$$

Mode:

Mode = none (all values are unique / no repeats)

2. Dispersion

$$\begin{aligned}\text{Range} &= \text{Max} - \text{Min} \\ &= 95 - 45 = 50\end{aligned}$$

Squared deviations from mean (65.7):

$$428.5, 246.5, 114.5, 32.5, 13.7, 0.5, 18.5, 86.5, 204.5, 858.5$$

$$\sigma = \sqrt{(\Sigma(x-\bar{x})^2/n)}$$

$$\text{Sum of squares} = 2004.1$$

$$\text{Variance } (\sigma^2) = 2004.1/10 = 200.41$$

$$\text{Standard deviation } \sigma = \sqrt{200.41} \approx 14.16$$

3. Quartiles & IQR

Lower half:

$$45, 50, 55, 60, 62$$

$$Q1 = 55 \text{ (middle value)}$$

Upper half:

65,70,75,80,95

Q3 = 75 (middle value)

$$\text{IQR} = Q3 - Q1 = 75 - 55 = 20$$

4. Outlier Check (IQR method)

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR} = 55 - 30 = 25$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR} = 75 + 30 = 105$$

All values lie within [25, 105] → no outliers detected (95 is high but still within bound)

5. Skewness

$$\text{Skewness} = 3 \times (\text{Mean} - \text{Median}) / \sigma$$

Skewness:

$$3 \times (65.7 - 63.5) / 14.16$$

$$= 3 \times 2.2 / 14.16 \approx 0.47$$

Mean (65.7) slightly greater than Median (63.5) → mild right (positive) skew

Summary Table

Metric	Value
Mean	65.7
Median	63.5
Mode	None
Range	50
Q1	55
Q3	75

IQR	20
Std. Deviation	≈ 14.16
Outliers	None
Skewness	$\approx +0.47$ (mild positive)

Overall Interpretation

- Students' marks are centered around 64–66 with moderate spread ($\sigma \approx 14.16$)
- No mode exists — every student scored a unique mark
- No true outliers, though the top score (95) pulls the mean slightly above the median
- The mild positive skew suggests a few higher scorers stretch the right tail of the distribution, while most students cluster in the mid-range

Answer: The dataset is approximately symmetric with a mild positive skew, moderate spread, and no statistical outliers.